

Technology Stack

Date	10 July 2026
Team ID	xxxxxx
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	To create a responsive and user-friendly web interface.

2	Backend / Server-Side	Python(flask)	To process user requests and run the flood prediction model
3	Database / Data Storage	My SQL	To store user details, flood history, and prediction records.
4	Cloud / Hosting / Deployment	Google Colab/ GitHub	Google Colab is used for model execution, GitHub for project hosting and version sharing.
5	Version Control & CI/CD	Git & GitHub	To manage source code, collaboration, and version control.

6	Third-Party APIs / Other Tools	Pandas, Numpy, Scikit- learn,matplotlib	To manage source code, collaboration, and version control.